

Syed Athar Hussain

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SUMMARY

Software Engineer with 4 years of experience specialising in data provisioning, ingestion pipelines, and database optimisation. Skilled in Object-Oriented Python, application-level SQL development, and cross-system integration bridging Python and Java environments. Proven track record in the Finance and Healthcare domains, delivering scalable ETL solutions, robust data quality frameworks, and architectural modernisations within Agile environments. Willing to relocate to Donegal, Ireland.

TECHNICAL SKILLS

- **Languages & Paradigms:** Python, SQL, Java.
- **Relational Databases:** PostgreSQL, Azure SQL (SQL Server), MySQL.
- **Data Engineering & Pipelines:** Apache Kafka, Apache Spark, Pandas, Airflow.
- **Infrastructure & Tools:** Docker, Kubernetes, AWS, Git, CI/CD, BigQuery.
- **Methodologies:** Agile Delivery, Root-Cause Analysis, Data Modeling, Data Quality Validation.

EDUCATION

University College Dublin, Dublin, Ireland **Sep 2024 – Aug 2025**
Master in Computer Science (Negotiated Learning)

- **Key Modules:** Machine Learning, Data Science, Generative AI, Cloud Computing, Data Visualization, Big Data.

St. Mary's Technical Campus, Kolkata, India **Aug 2015 – Jul 2019**
Bachelor of Technology in Computer Science

- CGPA: 7.61; Key Coursework: AI, Computer Networks, Data Management.
- Developed an e-commerce application using Laravel, emphasising scalable backend services and user-focused features.

WORK EXPERIENCE

Fujitsu Consulting India, Pune, India **Feb 2022 – Aug 2024**
Assistant Application Developer

- Built and maintained Python, Spark, and Kafka pipelines to process and serve streaming event data to downstream applications.
- Automated data transformation tasks using Python, applying OOP principles to keep the codebase modular and reduce pipeline downtime.
- Supported on-prem and cloud data pipelines, performing root-cause analysis on data drift and implementing robust data quality checks to ensure accuracy.
- Optimised backend containerization using Docker and Kubernetes, configuring autoscaling to guarantee 99.9% uptime for reporting systems and reducing latency by 80%.
- Operated within an Agile delivery environment, collaborating with product managers to translate business KPIs into optimised backend architectures.

Avant-Grade Technologies, Kolkata, India **Mar 2019 – Sep 2021**
Junior Full-Stack Developer | Full-Stack Developer

- Optimised PostgreSQL schemas and implemented advanced indexing strategies, executing complex SQL queries that reduced database load and cut execution times by 75%.
- Collaborated directly with application teams working in Java-based systems, building RESTful services in Spring Boot to bridge backend data provisioning with frontend interfaces.
- Integrated event-driven architecture using Apache Kafka to enable fault-tolerant, high-availability inter-service communication.
- Participated in daily code reviews and Agile sprint planning to drive architectural modernization and continuous improvement initiatives.

PROJECTS

U.S. Stock Options Data Pipeline:

- Tech Stack: Object-Oriented Python, Apache Kafka, Azure SQL, InfluxDB
- Architected a highly available ETL engine to ingest, transform, and provision real-time stock market data for 120 individual ticker symbols in parallel.
- Wrote Python validation scripts to handle deduplication, type casting, and null values before loading financial data into the database.
- Split data storage pipelines, routing structured historical records to Azure SQL and real-time metric data to InfluxDB.

- Maintained a 99% HTTP 200 response rate under heavy load, ensuring stable financial data streams for enterprise reporting.
- Integrated comprehensive system telemetry, improving infrastructure visibility by 70% to proactively prevent performance bottlenecks.

Data Preprocessing Pipeline:

- Tech Stack: Python, Pandas, InfluxDB, API Integration, Memory-Mapped Arrays
- Built automated data fetching pipelines integrating real-time API sources (InfluxDB) and static CSVs to support predictive analytics.
- Programmed complex data transformations, including forward-fill resampling of asymmetric timestamps to standardised 1-minute intervals across 120 data streams.
- Engineered over 600 concurrent feature columns utilizing Python and Pandas for advanced statistical and momentum calculations.
- Optimised system memory constraints during large-scale data processing by implementing NumPy memory-mapped arrays, bypassing RAM limitations.
- Implemented automated data quality filters to drop invalid, empty, or static sequences prior to downstream consumption.

Healthcare Data Quality & Analytics Pipeline:

- Tech Stack: Python, SQL, Pandas, Google BigQuery, Looker Studio
- Developed a Python-based ingestion script to load and process raw, semi-structured healthcare records into a normalised relational format.
- Cleaned raw healthcare data using Pandas, writing scripts to handle missing values and standardize inconsistent categories.
- Transformed transactional timestamps into actionable metrics (e.g., Duration of Stay) using application-level SQL and Pandas calculations.
- Integrated the cleansed dataset into Google BigQuery to support high-performance querying and downstream analytics.
- Modeled relationships between patient demographics and billing profiles to provision transparent, self-service data views for hospital leadership.

AWARDS

- **UCD Advantage Award:** Recognised by University College Dublin for participation in co-curricular and community activities.
- **STAR Silver Awards:** Earned twice at Fujitsu for outstanding performance in system reliability and optimisations.
- Secured second position in Science Exhibition organised by National Service Scheme (NSS).
- Awarded First Position in the NSS Wall Magazine competition for the conceptualization and designed a visual narrative focused on global warming and environmental advocacy.

CERTIFICATION & EXTRA-CURRICULAR ACTIVITIES

- **Certifications:**
 - Full Stack Java Development
 - Data Science Foundation
 - Google SRE Fundamentals
- **Participations :**
 - GeeksforGeeks Hackathon: Competed in coding challenges, applying ML and backend skills.
 - HackeRank Netsoc 2024 Contest: Participated in problem-solving events focused on algorithms and data structures.